

InsightForge — Business Requirements Document

Project: InsightForge — Executive Analytics Platform

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1. Executive Summary

InsightForge is a configuration-driven executive analytics platform that converts verified business metrics into decision-ready narrative reports. The platform separates analytical computation from language generation: SQL and Python calculate every metric, status classification, and diagnostic ranking, while a language model is limited to communicating those pre-verified findings in executive language. No metric, threshold, or business conclusion originates from the language model itself.

The platform has been implemented and validated against two independent business domains — Retail Sales Analytics and Telecom Service Assurance Analytics — using the same core engine and a domain-specific configuration file for each. This demonstrates that the underlying architecture generalizes across business functions rather than being built for a single dataset and that new domain can be supported through configuration rather than through changes to the analytical or narration code.

This document defines the business requirements for InsightForge: the problem it addresses, the scope of the current implementation, the functional and non-functional requirements it satisfies, the business rules governing how findings are calculated and communicated, and the constraints and risks associated with the current design.

2. Business Background

Executive and operational reporting in most organizations depends on analysts manually converting raw transactional and operational data into narrative summaries for leadership. This process is time-consuming, varies between analysts, and is difficult to audit after the fact, since the reasoning behind a stated conclusion is rarely captured alongside the figure itself.

Separately, the growing use of language models for business reporting has introduced a different risk. These models produce fluent narrative summaries, but without controls they can also generate plausible figures, trends, or conclusions that the underlying data does not actually support. In executive reporting, an unverifiable number is often more damaging than no number at all, because it can be acted on before it is questioned.

InsightForge's architecture is built against this background. It assumes that trust in an executive report depends on every figure being traceable to a deterministic calculation, and that a language model's role should be limited to communication rather than analysis. The platform was implemented and proven against a retail sales dataset and a telecom service assurance dataset, testing early on whether the same deterministic-first approach could generalize across unrelated business domains.

3. Business Problem Statement

Converting verified analytical results into executive-ready reporting currently requires manual analyst effort for every reporting cycle, and that effort does not transfer cleanly across business domains — a report format built for retail sales does not directly apply to telecom service assurance, or to any other function, without substantial rework.

At the same time, applying a language model directly to raw or aggregated data risks introducing unverified figures, invented trends, or causal claims the underlying data does not support. This is particularly damaging in executive reporting, where a single unsupported claim can undermine confidence in an entire report.

The business problem InsightForge addresses is the absence of a reporting approach that is efficient to extend across domains while remaining strictly bound to verified analytical evidence.

4. Business Objectives

- Convert verified, deterministic analytics into clear executive narrative, with no analytical responsibility assigned to the language model.
- Support additional business domains through configuration rather than through changes to core analytical or narration logic, demonstrated using two independent datasets.
- Preserve a direct, traceable link between every reported figure and its deterministic source.
- Provide both a structured executive report and an interactive question-and-answer mode over the same verified findings.
- Maintain consistent evidence and recommendation discipline across domains, so findings are never overstated relative to the underlying analytics.
- Ensure consistent executive reporting regardless of business domain by enforcing a common deterministic analytics workflow.

5. High-Level Business Process

InsightForge follows a deterministic analytics workflow in which each layer performs a single, well-defined responsibility before passing verified results to the next stage. Business metrics are calculated, interpreted, and validated before executive narration begins, ensuring that every reported finding remains explainable and traceable.

Step	Business Process	Outcome
1	Business Data	Operational and transactional source data
2	SQL Analytics	Data preparation and KPI aggregation
3	Python Analytics Engine	Deterministic KPI calculation
4	Business Interpretation Layer	Status classification, diagnostics, and comparisons
5	Executive Narration Layer	Executive communication of verified findings only
6	Executive Report & Interactive Analyst	Executive report generation and natural language business Q&A

6. Project Scope

6.1 In Scope

- Deterministic KPI calculation using SQL aggregation and Python computation, covering the eight retail metrics (Sales, Profit, Profit Margin, Average Order Value, Orders, Quantity, Average Shipping Days, Shipping Cost) and seven telecom metrics (Incident Count, SLA Breach Rate, Average Resolution Time, Total Cost, Cost per Incident, Customers Impacted, Dispatch Cost) defined in each domain's configuration.
- Deterministic business interpretation, including status classification where thresholds are configured, and descriptive reporting where they are not.

- Deterministic diagnostic ranking of contributors — for example, top sales market, top profit category, bottom profit subcategory, and top shipping cost ship mode for retail, or top states, top vendors, and top fault categories for telecom.
- Deterministic period-over-period comparison, where explicitly configured for a given metric.
- Executive narration of verified findings by a language model, governed by an explicit set of allowed and disallowed behaviors.
- A structured executive report and an interactive, single-metric question-and-answer mode.
- Configuration-driven extension to additional business domains through YAML configuration, without modifying the core engine.

6.2 Out of Scope

- Any metric calculation, KPI derivation, or business rule performed by the language model.
- Forecasting, predictive modeling, or machine learning of any kind.
- Root-cause determination beyond what deterministic diagnostics identify as a top or bottom contributor.
- Interactive dashboards or data visualization interfaces — identified as a future direction, not part of the current implementation.
- Cloud hosting or web-based deployment; the current implementation runs as a local Python application against a configured MySQL database.
- Automated corrective actions or staffing or budget changes generated from report findings.

7. Functional Requirements

ID	Category	Requirement
FR-1	KPI Engine	Calculate all configured KPIs via SQL aggregation against each project's configured database view (for example, vw_sales_flat for the retail configuration).
FR-2	KPI Engine	Compute calculated KPIs, such as profit margin and average order value, from configuration-defined formulas rather than hard-coded logic.
FR-3	KPI Engine	Classify a KPI's status as Good, Warning, or Critical only when thresholds are defined in configuration; report all other KPIs as Descriptive.
FR-4	Interpretation	Generate a deterministic business meaning and interpretation for every calculated KPI.
FR-5	Interpretation	Generate deterministic diagnostic rankings for configured dimensions, such as market, category, subcategory, and ship mode.
FR-6	Interpretation	Generate a period-over-period comparison — window, current value, previous value, percentage change, and direction — only where comparison logic is explicitly configured for a KPI.
FR-7	Interpretation	Raise an explicit error when no interpretation rule exists for a requested metric, rather than returning an unverified result.
FR-8	Narration	Generate executive narrative exclusively from findings supplied by the KPI Engine and Business Interpretation Layer.
FR-9	Narration	Report comparisons using only the supplied direction label (Increase, Decrease, or Stable); never substitute inferred qualitative language.

ID	Category	Requirement
FR-10	Narration	Present diagnostics as a structured table using supplied display values only, without duplicating the same ranking in surrounding prose.
FR-11	Narration	Prioritize material negative findings and verified diagnostics ahead of routine descriptive metrics in the executive summary.
FR-12	Narration	Withhold the full executive report, and state that insufficient verified metrics are available, when fewer than 50% of requested metrics are successfully supplied.
FR-13	Reporting	Generate a structured executive report containing an executive summary, operational performance, financial performance, business implications, and recommendations, with support for export as a formatted document.
FR-14	Reporting	Support an interactive mode that returns a plain-language answer to a single-metric question, without full report formatting.
FR-15	Reporting	Generate recommendations using only an approved set of verbs — investigate, review, monitor, compare, establish baseline, continue monitoring — aligned to the specific metric each diagnostic represents.
FR-16	Configuration	Load all domain-specific settings — database view, dimensions, measures, KPI definitions, thresholds, and business assumptions such as currency — from an external YAML configuration file.
FR-17	Configuration	Support multiple independently configured business domains using the same core engine, without modifying the analytics engine or executive narration layer.

8. Non-Functional Requirements

- **Explainability** — every reported figure must be traceable to a deterministic SQL or Python calculation; no figure may originate from the language model.
- **Consistency** — the same metric, requested through the executive report or the interactive mode, must return the same underlying value and interpretation.
- **Audit ability** — business meaning, interpretation, and threshold logic must be defined in code and configuration, not inferred at generation time.
- **Extensibility** — adding a new business domain must require only a new configuration file and, where needed, new interpretation rules, not a redesign of the KPI engine or narration layer.
- **Data integrity** — numerical values passed to the narration layer must be presented exactly as supplied, without recalculation, rounding changes, or unit or currency conversion.
- **Credential handling** — database and API credentials are managed through environment variables and excluded from version control.

9. Business Rules

ID	Business Rule
BR-1	The narration layer may improve readability and structure but may never calculate a metric, write a query, or derive

ID	Business Rule
	a KPI.
BR-2	Deterministic status and direction labels must be used exactly as supplied; the narration layer may not substitute synonymous language.
BR-3	A metric with no supplied comparison must be reported descriptively; trend language may not be applied to a single reporting period.
BR-4	A reporting period may not be described as a baseline or benchmark unless deterministic analytics explicitly designates it as such.
BR-5	Co-occurring findings may be reported together, but a causal or predictive relationship may only be stated if deterministic analytics explicitly supplies it.
BR-6	Diagnostic rankings identify top or bottom contributors only; recommendations may not imply that the top contributor represents poor performance.
BR-7	Recommendations must align with the specific metric each diagnostic represents, and may not include optimization, automation, staffing, or budget changes unless deterministic findings explicitly justify them.
BR-8	A reporting limitation shared across multiple metrics is stated once, in the earliest relevant section, rather than repeated throughout the report.

10. Solution Overview

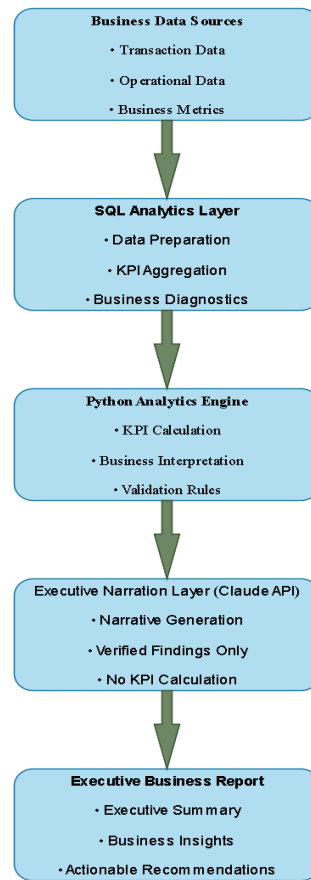
InsightForge is structured as a pipeline moving from raw data to executive communication, with each stage owning a single responsibility: SQL Analytics, a Deterministic KPI Engine, a Business Interpretation Layer, and an Executive Narration Layer built on Claude.

SQL performs all aggregation against a configured database view. The KPI Engine computes each defined metric — either as a direct aggregation or a configured formula — and acts as the single source of truth for reported values. The Business Interpretation Layer applies deterministic rules to classify status where thresholds exist, generate diagnostic rankings, and produce comparison data where configured. Only after a finding has passed through these layers does it reach the Executive Narration Layer, which converts it into executive language without altering its meaning or strength.

The same pipeline supports both implemented domains, and the difference between them illustrates how the platform is intended to scale. For Retail Sales Analytics, no thresholds are configured, so all findings are reported descriptively. For Telecom Service Assurance Analytics, at least one metric — SLA breach rate — has configured thresholds and a period-over-period comparison, allowing the same engine to produce a Good, Warning, or Critical classification and a trend direction where the retail configuration cannot. This difference is a product of configuration, not of separate code paths.

11. System Architecture

InsightForge System Architecture



Component	Technology	Responsibility
Database	MySQL (via SQLAlchemy, PyMySQL)	Stores source data; exposes analytical views
KPI Engine	Python	Deterministic KPI computation
Business Interpretation Layer	Python	Status classification, diagnostics, comparisons
Executive Narration Layer	Claude, governed by a defined system prompt	Converts verified findings into executive language
Configuration	YAML (via PyYAML)	Domain-specific KPIs, thresholds, dimensions, business assumptions
Report Output	python-docx, reportlab	Formatted document generation
Application Entry Point	main.py	Generates the executive report and launches the interactive analyst

The repository separates these responsibilities into dedicated directories: `agent/` for narration and report generation, `engine/` for KPI calculation and business interpretation, `config/` for YAML configuration, `sql/` for schema and analytical queries, and `docs/` for project documentation, including this document.

12. User Workflow

InsightForge supports two workflows over the same verified findings.

Executive report: The user runs the application against a configured project. The system loads the relevant configuration, executes the configured SQL queries, and computes all defined KPIs. The Business Interpretation Layer classifies and diagnoses each finding, and the Executive Narration Layer generates a structured report: an executive KPI scorecard, followed by executive summary, operational performance, financial performance, business implications, and recommendations sections.

Interactive analyst: The user launches the interactive mode from the same application and asks a single question in natural language — for example, "Which market generated the highest sales?" The system retrieves the relevant verified finding and returns a plain-language answer, without full report formatting. If a question requires evidence that has not been supplied, such as a year-over-year comparison that was not configured, the system states that the comparison cannot be determined rather than estimating an answer.

13. Assumptions

- The Business Background and Problem Statement in Sections 2 and 3 are derived from InsightForge's documented design principles and evidence-first architecture; no separate stakeholder-supplied problem statement was provided as source material.
- Source data is assumed to already exist in each project's configured database view prior to report generation; the process that populates this view is not documented and is treated as outside the current implementation.
- Currency and unit conventions, such as INR for the retail dataset, are set per project configuration and are not assumed to be universal across domains.
- Database and language model API credentials are assumed to be provisioned and available via environment variables prior to execution.
- Source documentation describes the two validated datasets differently: the platform overview identifies the telecom dataset as the flagship implementation, while the retail configuration file's internal description labels itself as the flagship configuration. This document treats both as equally validated implementations of the same core engine, consistent with how they are described in the original project brief, rather than resolving the discrepancy in favor of one dataset.

14. Constraints

- The narration layer can only report what the deterministic layers supply; it cannot answer questions or generate findings for metrics, comparisons, or dimensions outside the current configuration.
- Status classification is available only for KPIs with explicitly configured thresholds; KPIs without thresholds remain descriptive regardless of business importance.
- The current implementation is a local, database-connected Python application; it does not include a web interface, hosted deployment, or dashboard.
- Report generation depends on the availability of the configured database and the language model API; the system does not define a fallback beyond withholding the full report when metric coverage falls below 50%.

15. Risks

- Configuration errors in a project's YAML file, such as an incorrect column mapping or formula, would propagate directly into reported KPI values, since the KPI Engine is the sole source of truth and no independent validation layer is documented.
- Domains without configured thresholds or comparison logic, as with the current retail configuration, are limited to descriptive reporting, which reduces decision-support value until thresholds are established.
- The evidence boundary between deterministic analytics and narrative generation depends on the completeness of the rules supplied to the narration layer; extending the platform to a new domain without equally rigorous interpretation rules could weaken that boundary in practice, even though the architecture is designed to prevent it.
- The platform depends on external services — the configured database and the language model API — for both report generation and interactive use; unavailability of either would prevent the system from producing output.

16. Success Criteria

- Every figure in a generated report can be traced back to a specific SQL or Python calculation, with no figure originating from the narration layer.
- The same core engine and narration layer produce accurate, domain-appropriate reports for both the Retail Sales Analytics and Telecom Service Assurance Analytics configurations without code changes.
- Status classifications, trend directions, and diagnostic rankings in generated reports match the deterministic values and labels supplied by the Business Interpretation Layer, with no substituted or inferred language.
- Interactive queries that fall outside available evidence, such as an unconfigured comparison, are answered by stating that the evidence is unavailable, rather than by estimating a plausible answer.
- A new business domain can be on boarded by adding a YAML configuration file and, where necessary, corresponding interpretation rules, without modifying the KPI engine or narration layer.
- Interactive responses remain consistent with executive reports because both consume the same verified analytical findings.

17. Future Scalability

InsightForge is designed around a configuration-driven architecture rather than a domain-specific solution. KPI definitions, thresholds, dimensions, business rules, and interpretation logic are maintained through external configuration, allowing the same analytics framework to be reused across different business domains.

This approach has already been validated using two independent implementations—Retail Sales Analytics and Telecom Service Assurance Analytics—without changing the core SQL Analytics Layer, Python Analytics Engine, or Executive Narration Layer.

Future enhancements will build on the existing architecture rather than replace it. These include broader period-over-period analysis, additional deterministic KPI and diagnostic libraries, support for more business domains through configuration, and integration with interactive dashboards. These enhancements extend the platform's analytical capability while preserving the deterministic and explainable design established in the current implementation.